A View From Somewhere: Human-Centric Face Representations

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Motivation

Dataset Bias

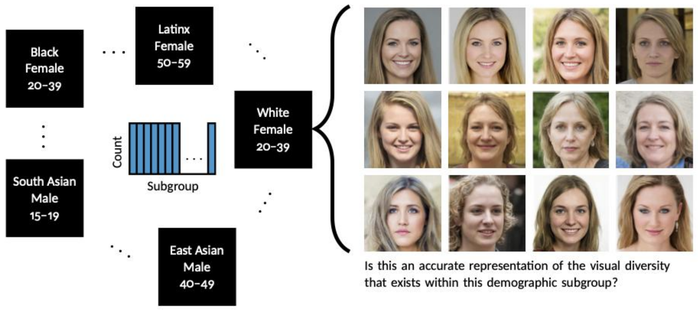
Dataset bias can result in models that are discriminatory or depend on spurious correlations. For instance, a face image restoration model may "hallucinate" features correlated with oversampled subgroups, resulting in the erasure of minority groups.

Original images are taken from the Chicago Face Database (https://www.chicagofaces.org)

Prior to learning, human-centric image dataset users should **evaluate the diversity of the data**.

Parity and Diversity

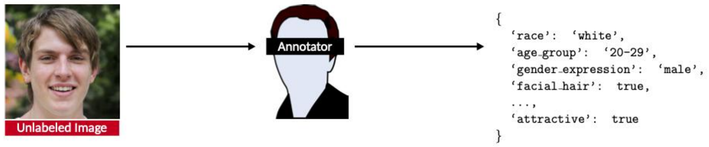
The canonical approach is to categorize people using demographic labels. However, evaluating diversity by examining counts across subgroups **fails to reflect the continuous nature of human phenotypic diversity** (e.g., skin tone is often reduced to `light` vs. `dark`). Moreover, such approaches often **deny multi-group membership** (e.g., erasing multi-ethnic individuals).



Is this an accurate representation of the visual diversity that exists within this demographic subgroup?

Unknown Label Distributions

Most human-centric image datasets are web-scraped, lacking ground-truth information about the image subjects. Therefore, researchers typically choose certain attributes they consider to be relevant for human diversity and use human annotators to infer them. This is difficult for ill-defined and highly changeable social constructs such as race and gender. **Observational labels risk not only encoding stereotypes, but reifying and propagating them beyond "their cultural context"**. Furthermore, discrepancies between observed and self-identified attributes can **invalidate an image subject's self-image and identity**.

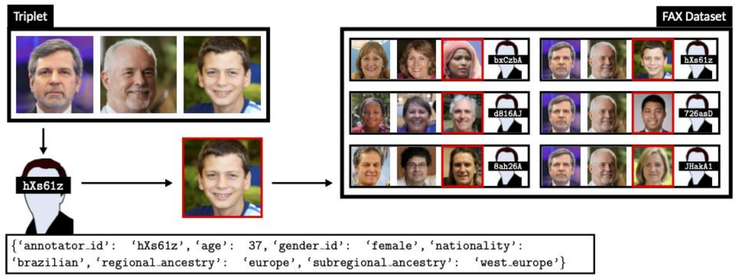


Contributions

Motivated by issues inherent to categorical labels, we aim to develop a tool that can evaluate the **visual diversity** of faces in unlabeled datasets. We do so without ever asking an annotator to explicitly categorize a person.

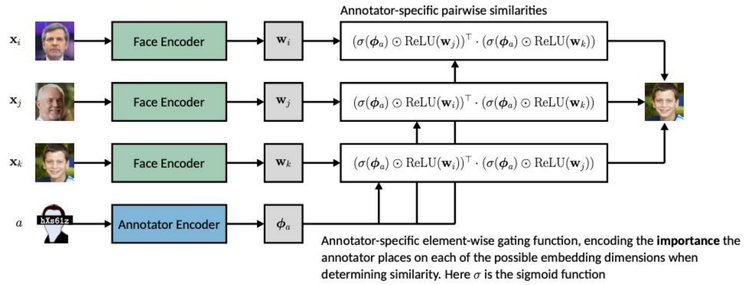
Face Similarity (FAX) Dataset

We introduce a novel dataset, `FAX`, containing 638,180 odd-one-out face similarity judgments over 4,921 faces (stimulus set). Judgments correspond to the odd-one-out (i.e., least similar) face in a triplet, and are associated with the identifier and demographic attributes of the annotator who made the judgment.



Model of Conditional Decision-Making

We introduce a model that learns to predict human judgments of face similarity conditioned on the annotator who generated the judgment. The conditions are realized using annotator masks.



A face can exhibit certain characteristics to a greater or lesser extent than others, even within the same subpopulation. Therefore, we constrain our model to learn face embeddings that are **continuous**, **non-negative (human-interpretable)**, and **sparse** by minimizing:

$$\underbrace{-\sum_n \log\left[\hat{p}(k \mid a)\right]}_{\text{Classification loss}} + \underbrace{\lambda_1 \sum_i \|\mathrm{ReLU}(\mathbf{w}_i)\|_1}_{\text{Sparsity loss}} + \underbrace{\lambda_2 \sum_{ij} \mathrm{ReLU}(-\mathbf{w}_i)_j}_{\text{Non-negative loss}} + \underbrace{\lambda_3 \sum_a \|\boldsymbol{\phi}_a\|_2^2}_{\text{Annotator mask loss}},$$

where $\hat{p}(k \mid a)$ is the predicted probability that face $\mathbf{x}_k$ is the odd-one-out according to annotator a.

Results

Interpretability

From learning on FAX, we note the materialization of distinct dimensions coinciding with commonly defined **demographic subgroups**, i.e., `Male`, `Female`, `Black`, `White`, `East Asian`, `South Asian`, and `Elderly`. In addition, separate dimensions surface for **face and hair morphology**, i.e., `Wide Face`, `Long Face`, `Smiling Expression`, `Neutral Expression`, `Balding`, `Facial Hair`, and `Dyed Hair`.

Original images are taken from FFHQ (https://github.com/NVlabs/ffhq-dataset)

Moreover, we validate that `FAX` dimensions can be used to **directly collect continuous attribute values for novel faces**, sidestepping the limits of categorical definitions.

Predicting Novel Similarity Judgments

Model	Loss/Method	Acc.	r
`FairFace`	Cross-entropy	51.9	0.67
`CelebA`	Cross-entropy	48.9	0.56
`FFHQ`	Cross-entropy	51.8	0.68
`CASIA-WebFace`	ArcFace	46.1	0.40
`FAX-C`	Conditional	**61.7**	0.82
`FAX-U`	Unconditional	57.5	**0.86**
`FAX-Triplet`	Triplet margin with distance swap	52.8	0.64

From 56 images, we generate $\binom{56}{3}$ possible triplets with 2–3 unique judgments per triplet (80,300 judgments).
We report odd-one-out predictive accuracy; and, Spearman's r between the strictly upper triangular model- and human-generated similarity matrices. Entry (i, j) in the human-generated similarity matrix corresponds to the fraction of triplets containing (i, j), where neither was judged as the odd-one-out. Entry (i, j) in a model-generated similarity matrix corresponds to the mean $\hat{p}(k)$ over all triplets containing (i, j).

Annotator Bias

Annotator attribute	Groups	# Masks	AUROC
Age group	30–39 / 40–49	393 / 121	0.59 ± 0.05
Gender identity	`Male` / `Female`	523 / 473	0.65 ± 0.05
Nationality	`America` / `India`	530 / 204	0.86 ± 0.03
Regional ancestry	`Europe` / `Asia`	407 / 243	0.86 ± 0.03
Subregional ancestry	`West Europe` / `South Asia`	173 / 107	0.88 ± 0.05

Using 10-fold cross validation, we train linear SVMs to discriminate between annotator masks from two different annotator subgroups.

Comparative Diversity Auditing and Binary Attribute Classification

Comparative diversity auditing

		Model (Disparity Δ / Spearman's r)			
Data	Attribute	`FAX`	`CelebA`	`FairFace`	`FFHQ`
CC	> 70 y.o.*	0.22 / 0.96	**0.06** / **0.99**	**0.06** / 0.95	0.08 / **0.99**
CC	Male*	0.06 / 0.97	**0.02** / 0.99	**0.02** / **1.00**	0.04 / 0.99
CC	Light skin	**0.06** / 0.94	0.3 / 0.65	0.18 / 0.85	**0.06** / **0.99**
CFD	Smiling	0.12 / **0.99**	0.06 / **0.95**	—	—
CFD	Male*	**0.00** / **1.00**	0.08 / 0.99	0.02 / **1.00**	0.02 / **1.00**
CFD	East Asian*	0.06 / **1.00**	—	**0.02** / 0.98	**0.02** / **1.00**
CFD	Black*	0.10 / 0.98	—	0.14 / 0.97	**0.00** / **1.00**
CFD	White*	0.04 / **1.00**	**0.02** / 0.99	0.04 / 0.97	0.04 / 0.99
CFD	Indian*	**0.08** / **1.00**	—	0.30 / 0.73	**0.08** / 0.87

Given a set of candidate subsets, we aim to find the most diverse subset (i.e., minimal attribute disparity).
`FAX` dimensions result in the identification of data subsets with competitive attribute disparity scores. In particular, Spearman's r evidences that `FAX` disparity scores are highly correlated with the ground-truth label disparity.

Discriminating between binary-valued attributes

		AUROC			
Data	Attribute	`FAX`	`CelebA`	`FairFace`	`FFHQ`
CC	> 70 y.o.*	0.800	0.936	0.959	0.962
FFHQ	> 70 y.o.	0.905	0.959	**0.971**	0.980
CFD	Male*	0.991	0.997	0.996	**0.998**
CC	Male*	0.971	0.986	**0.990**	0.981
CA	Male	0.990	0.999	0.994	**0.995**
COCO	Male	0.893	0.926	**0.963**	0.958
MIAP	Male	0.924	0.942	**0.945**	0.938
FFHQ	Male	0.933	0.959	**0.988**	0.996
CA	Smiling	**0.895**	0.982	—	—
CFD	Smiling	0.969	**0.992**	—	—
CFD	Neutral	**0.731**	—	—	—
CFD	East Asian*	**0.969**	—	0.955	0.938
CFD	Black*	**0.992**	—	**0.992**	0.988
CFD	White*	0.972	0.889	**0.990**	0.988
CFD	Indian*	**0.960**	—	0.848	0.883
CC	Light skin	0.930	0.830	**0.965**	0.960
COCO	Light skin	0.889	0.771	0.927	**0.931**
CA	Balding	**0.963**	0.995	—	—

`FAX` dimensions are competitive with the baselines, even in challenging unconstrained settings as represented by COCO and MIAP.